

# ADITYA TIWARI

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## Profile Summary

Results-oriented ML & Platform Engineer with nearly 4 years of experience delivering highly reliable AI solutions and overseeing robust infrastructure operations. I specialize in bridging advanced predictive modeling with rigorous platform management, ensuring machine learning systems are seamlessly deployed, scaled, and continuously maintained. My expertise spans the complete AI lifecycle, from driving zero-downtime platform migrations and ensuring cluster reliability, to architecting next-generation Generative AI solutions, including sophisticated Retrieval-Augmented Generation (RAG) systems. I am dedicated to championing operational stability and production excellence for complex, data-driven applications.

## Experience

### Data Scientist

Jio Platforms Limited

Mumbai, India

July 2022 – Present

- **End-to-End Platform Ownership:** Manage the internal JDSP Platform for the organization, taking full responsibility for development, cluster management, and maintaining a high-availability environment 99.9% for 500+ users.
- **Infrastructure Migration & Stabilization:** Successfully executed a zero-downtime migration of legacy Airflow and JDSP workloads, stabilizing core infrastructure and leading complex user workspace transitions across environments (e.g., HDP to CDP).
- **Containerization & Orchestration:** Orchestrate containerized services using Kubernetes and Docker, managing liveness probes and troubleshooting complex pod lifecycle issues (Exit Code 137) to ensure system resilience.
- **Systems & OS Management:** Handle comprehensive on-premise server activities, including OS-level maintenance, cluster upgrades, scaling operations, and infrastructure optimization to support data-intensive workloads.
- **Automation & CI/CD:** Streamline operational efficiency by automating manual processes and maintaining robust deployment pipelines through Git/GitHub, ensuring reliable version control and platform growth.
- **Expert Technical Support:** Provide dedicated L1/L2 end-to-end support, troubleshooting platform-specific bottlenecks and delivering fast resolution for data engineering and data science teams.
- **Advanced Workflow Integration:** Lead feature enhancements to streamline workflows, integrating modern tools for data orchestration with Apache Airflow and preparing the platform for scalable AI and RAG-based application deployments.

## Projects

### Stateful AI Travel Agent (LangGraph, Gemini, Streamlit)

[Project Report](#) 

- Architected a stateful AI agent using LangGraph and LangChain, powered by the Gemini LLM, to autonomously reason and generate dynamic, personalized travel itineraries.
- Integrated the Tavily Search API as a custom tool within the agentic workflow, enabling real-time data retrieval and reducing AI hallucinations for location-specific queries.
- Developed an interactive front-end using Streamlit to facilitate seamless human-AI interaction, structuring the backend for clean, scalable local deployment.

## Technical Skills

- **Programming / Scripting:** Python, SQL, Bash Scripting
- **ML Frameworks:** PyTorch, TensorFlow
- **Generative AI & LLMs:** LangChain, Hugging Face, RAG, Gemini API, LangGraph
- **MLOps & Infrastructure:** Apache Airflow, Kubernetes (K8s), Docker, GitLab, Prometheus, Grafana
- **Data Analysis & Backend:** Pandas, NumPy, FastAPI, Streamlit

## Education

### Global Nature Care Sangathan's Group Of Institution, Jabalpur

BTech in Electronics And Telecommunication

July 2018 – July 2022

- CGPA: 8.35/10.0

## Certifications

- **Machine Learning Specialization** – Stanford Online / DeepLearning.AI (2025)
- **Generative AI with Large Language Models Specialization** – DeepLearning.AI (2026)
- **Python for Everybody Specialization** – University of Michigan (2024)